

Streamlining Faculty Support: BISU Faculty Manual Chatbot

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Abstract—Chatbots have changed the way information can be accessed through developments in natural language processing (NLP) and machine learning technologies. This study set up the BISU Faculty Manual Chatbot to solve the problems that faculty members encounter in navigating the faculty manual at Bohol Island State University – Clarin Campus, which is often considered ineffective because it is paper-based. It uses the TinyLlama generative artificial intelligence model, which has been fine-tuned with the BISU Faculty Manual 2022 Revised Edition to produce accurate and contextually appropriate responses. Django was used both to build the frontend and the backend, and the model, TinyLlama, was trained and fine-tuned in Google Colaboratory. The faculty manual data was prepared in advance, namely tokenized and labeled, to enhance the way the chatbot can understand user queries more effectively. Evaluation metrics were considered such as Fallback Rate, Handoff Rate, and Containment Rate to determine the performance of the chatbot. The Containment Rate of the chatbot was 86.3%, the Fallback Rate was 6.1%, and the Handoff Rate was 7.6%, thereby indicating that it can solve user queries on its own with minimal errors. A customer satisfaction rating of 4 stars reflects its design effectiveness and reliability. In conclusion, the BISU Faculty Manual Chatbot has numerous benefits: it reduces faculty workload, saves time, and increases accessibility. Recommendations include dataset expansion and conversational memory integration for enhanced performance.

Keywords—Chatbot, NLP, Faculty Support, Machine Learning, TinyLlama, Manual

I. INTRODUCTION

Faculty manuals are vital resources that provide institutional policies, procedures, and guidelines, helping faculty members navigate their roles effectively. According to Dr. Margie Avenido, President of the Faculty Association at Bohol Island State University-Clarin Campus, the faculty manual acts as a "bible" for faculty, outlining benefits and processes. However, traditional manuals pose challenges, such as inefficiency in locating information, lack of real-time accessibility, and time-consuming page scanning. Dr. Avenido emphasizes the need

for alternative methods to streamline access, suggesting the transformation of the manual into a chatbot.

The study "Streamlining Faculty Support: BISU Faculty Manual Chatbot" addresses these challenges by proposing an AI-based solution. The chatbot aims to enhance educational efficiency by providing instant access to information, reducing administrative burdens, and improving faculty performance and satisfaction. This initiative reflects BISU's commitment to leveraging technology for better educational outcomes and setting a benchmark for innovation in faculty support services.

II. RELATED WORKS

The literature background of this study highlights the relevance of faculty manual chatbots, supported by Republic Act No. 8292 (Higher Education Modernization Act of 1997) and Article XIV Section 10 of the Philippine Constitution [1] [2]. These legal frameworks emphasize the importance of leveraging technology to enhance teaching, learning, and equitable access to quality education. By incorporating the IPO model and synthesizing related studies, this research explores the design, functionalities, and potential impacts of faculty manual chatbots in academic settings.

Republic Act No. 8292 provides a legal foundation for adopting innovative technologies in education. Section 2 of the Act underscores the state's policy to create a comprehensive education system aligned with societal needs. This study aligns with the Act's goals by proposing a chatbot to replace traditional manual systems, offering faculty efficient access to information and enhancing institutional effectiveness.

Additionally, Article XIV Section 10 of the Philippine Constitution emphasizes the role of science and technology in national development. It calls for prioritizing research, innovation, and self-reliant technological capabilities. This constitutional mandate further justifies the study, as the faculty manual chatbot represents a technological advancement aimed

at improving educational systems and contributing to national progress.

This study adopts the Input-Process-Output (IPO) Model as its theoretical and conceptual framework. Widely used in system analysis and software engineering, the IPO model describes the structure of information processing systems. The IPO model encompasses all process components, including inputs, process details, resources, and outputs, providing a comprehensive representation of the system's workflow [3].

In [4], Wagner discusses Meta's release of the Llama 3 AI model, designed to enhance chatbot capabilities in reasoning, learning, and generating coherent responses. Llama 3 is integrated into Meta's platforms like Facebook and WhatsApp, and its open-source nature enables developers to build tailored chatbots, such as a faculty manual chatbot for higher education institutions. Additionally, [5] highlights the shift from traditional methods to deep learning models, particularly encoder-decoder architectures, for chatbot development. The study emphasizes the challenges of understanding human utterances and generating appropriate responses, proposing a functional architecture for intelligent conversational agents in healthcare.

In [6], Shah et al. developed an intelligent chatbot using natural language processing (NLP), emphasizing its potential to revolutionize educational systems by providing cost-effective and inclusive solutions. Similarly, [7] proposes the UNISEL Crawler system, a chatbot designed to address challenges like manual information dissemination and lack of real-time support in universities. Using the Agile approach, the system enhances customer service and user engagement through technology-driven solutions. Both studies highlight the relevance and effectiveness of NLP-based chatbots in addressing specific institutional needs.

In [8], Sadka discusses the advancements in OpenAI's GPT-5 and Meta's Llama-3, highlighting their enhanced capabilities in multilingual communication, contextual understanding, and emotional intelligence. These improvements align with the objectives of developing a faculty manual chatbot, enabling more natural and intelligent interactions. Similarly, [9] introduces TinyLlama, a compact 1.1-billion-parameter model pretrained on 1 trillion tokens, which outperforms larger models in specific tasks, demonstrating its potential for efficient language model development. Additionally, [10] emphasizes the role of chatbots in education, particularly in large classes, where they provide scalable, personalized support to complement human instructors.

This aligns with the study's objective of leveraging technology and NLP to improve accessibility and user experience in educational institutions. Both studies address challenges like manual information dissemination and the need for real-time support, reinforcing the rationale for developing a faculty manual chatbot to enhance management processes and provide immediate assistance to faculty members, meeting the evolving demands of educational environments.

In [11], Shivashankar et al. introduce EVA, a deep learning-based campus assistive chatbot that uses NLP and neural networks to provide concise information and navigation sup-

port, achieving an 89% performance rate. Similarly, [12] presents NEU-chatbot, an AI-powered system for admissions and curriculum updates at National Economics University, achieving 97.1% accuracy with a Vietnamese tokenizer. In [13], Patel et al. propose UNIBOT, a web-based university chatbot that leverages AI and PHP to provide seamless access to university-related information. Additionally, [14] explores a chatbot for educational support, utilizing NLP and ontologies to deliver accurate responses to student queries, demonstrating its effectiveness as an educational tool.

In [15], Lalwani et al. develop a chatbot using AI and NLP to simplify access to college-related information, addressing challenges in navigating institutional websites. Similarly, [16] proposes an NLP-based chatbot using PyTorch and NLTK, adaptable across domains like healthcare and e-commerce. In [17], Susanna et al. create a college inquiry chatbot to streamline information retrieval for students, enhancing user experience. Additionally, [18] introduces a smart college chatbot using ML and WordNet to improve interactions with a university website, while [19] presents an NLP-based chatbot framework for higher education, deployed on Heroku with plans for AI training and multi-language support.

The literature review reveals a common focus on leveraging NLP and ML techniques to develop chatbots for educational contexts. These studies highlight the role of advanced technologies in enhancing communication, addressing diverse queries, and supporting personalized learning. They emphasize the adaptability of chatbots and underscore the growing importance of AI-driven solutions in transforming educational interactions and improving information dissemination in higher education.

Thus, the thesis writers aim to address these gaps by developing more user-friendly web applications to assist the faculty members of BISU in accessing essential information and services seamlessly.

The thesis writers developed a model using advanced NLP algorithms to interpret user inquiries and deliver accurate responses. Designed for BISU faculty, the model streamlines access to essential information and services, ensuring efficiency in addressing academic and administrative needs. It provides a user-friendly interface for navigating policies and tasks seamlessly.

III. OBJECTIVE

The primary objective of this study is to develop and implement the BISU Faculty Manual Chatbot to enhance the accessibility of institutional policies and guidelines to faculty. The chatbot will eliminate the process of retrieving information by allowing faculty members to use a useful and effective tool that provides timely and accurate information to queries raised on the BISU Faculty Manual. Based on the TinyLlama generative AI model fine-tuned from the BISU Faculty Manual 2022 Revised Edition, the chatbot will be responsible for providing appropriate and context-related answers. Aside from that, this study aims to improve accessibility through the possibility of accessing information at any given time and place with an

interactive tool. To determine its effectiveness, the chatbot will be measured based on the key performance parameters of Containment Rate, Fallback Rate, Handoff Rate, and user satisfaction surveys. Ultimately, this study wants to enhance the efficiency of an institution by alleviating faculty load, reducing search time for policies, and making productivity better overall.

IV. METHODOLOGY

A. Dataset

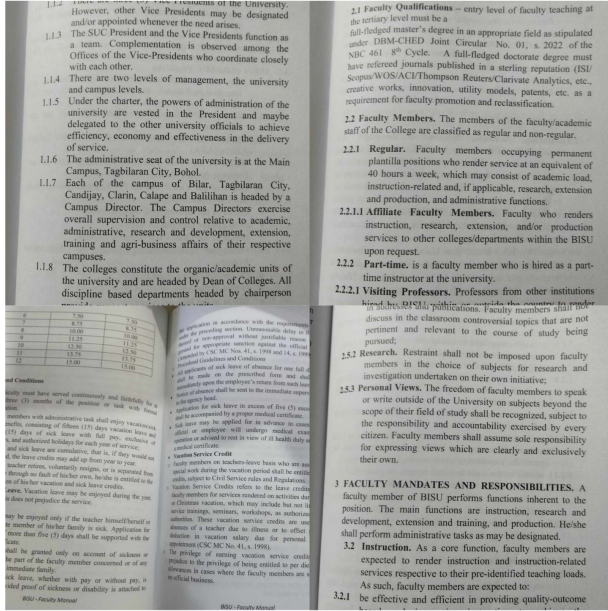


Fig. 1. (BISU Faculty Manual 2022 Revised Edition) Sample Dataset

Figure 1 shows the dataset utilized for this study, which is the Bohol Island State University Faculty Manual 2022 Revised Edition. This comprehensive manual serves as a vital resource for faculty members, offering detailed guidelines and regulations pertaining to various aspects of their roles and responsibilities within the university. It provides clarity on academic policies, administrative procedures, and other essential information for efficient faculty management. The dataset, structured and stored in a CSV file format, was used for fine-tuning the TinyLlama model. It contains various statements users might employ to inquire about faculty manual topics, enabling the TinyLlama model to learn and generate responses that accurately reflect the nuances of these inquiries.

B. System Architecture

The system architecture is the design and structure that outlines the seamless integration of its core components and ensures efficient functionality. It overviews the system's design, highlighting the connections between authentication, frontend, backend, database, and AI-powered chatbot services.

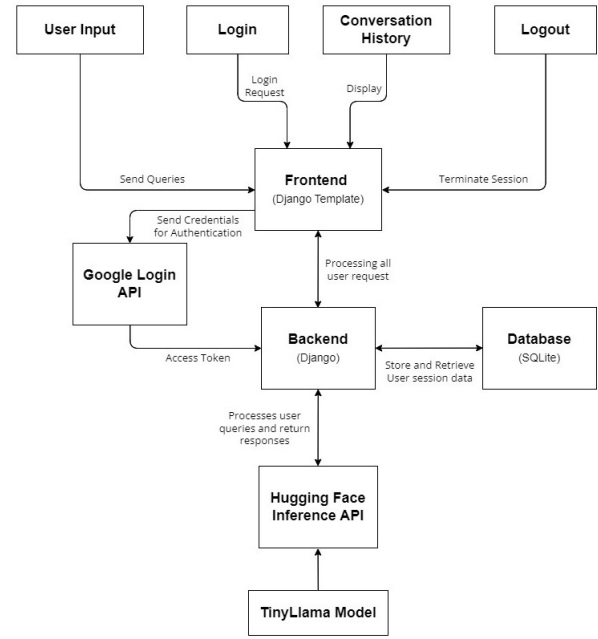


Fig. 2. System Architecture

Figure 2 shows the system's final architecture. The system architecture is designed to ensure seamless communication between its components while maintaining functionality and scalability. The Google Login API handles user authentication, allowing secure access to the system. The Django framework serves as both the frontend and backend, managing user inputs, processing chatbot queries, and communicating with the Hugging Face Inference API, which hosts the trained TinyLlama model for generating responses. Additionally, the SQLite database stores conversation history, enabling users to revisit past interactions, while the backend ensures efficient data flow between all components.

C. Environment and Training Parameters

TABLE I
HYPERPARAMETERS

Hyperparameters	Scores
Batch Size	4
Gradient Accumulation Steps	2
Optimizer	paged_adamw_32bit
Learning Rate	0.001
Logging Steps	10
Learning Rate Scheduler Type	cosine
Save Strategy	epoch
Epoch	40

Table I shows the environment and training parameters used in developing the BISU Faculty Manual Chatbot. The model was trained using Google Colaboratory, leveraging its cloud-based GPU support to expedite the fine-tuning process. The TinyLlama generative AI model was fine-tuned with the BISU Faculty Manual 2022 Revised Edition, ensuring that responses

were contextually relevant and aligned with institutional policies. The dataset underwent tokenization and preprocessing to improve model comprehension and accuracy. The training process utilized a batch size of 4 was set, with gradient accumulation steps of 2, simulating an effective batch size of 8. The optimizer, `paged_adamw_32bit`, ensured memory-efficient weight updates, while a learning rate of 0.001 controlled update sizes. A cosine learning rate scheduler improved convergence, and checkpoints were saved every epoch. Logs were generated every 10 steps for monitoring, and 40 epochs were fixed to optimize learning. These hyperparameters balanced computational efficiency and model performance, tailored for training the TinyLlama model.

D. Performance Measures

Table II presents the chatbot’s performance based on Fallback Rate, Handoff Rate, Containment Rate and Customer Satisfaction Rating as evaluated by faculty members. A random sampling method was used to select the faculty participants for system evaluation. The chatbot’s performance metrics were calculated from a total of 43 user interactions, with the results detailed below.

TABLE II
PERFORMANCE OF THE CHATBOT

Metrics	Results
Fallback Rate	6.1%
Handoff Rate	7.6%
Containment Rate	86.3%
Customer Satisfaction Rating	4 stars

Table II outlines the chatbot’s performance metrics, demonstrating its effectiveness in handling user interactions. The chatbot achieved a Fallback Rate of 6.1%, indicating it failed to understand or respond accurately in only 6.1% of interactions, and a Handoff Rate of 7.6%, reflecting minimal reliance on human intervention. These results align with the study’s objective of minimizing both metrics, as a lower Fallback Rate signifies stronger independent query resolution, and a lower Handoff Rate highlights conversational competence. Additionally, the chatbot achieved a Containment Rate of 86.3%, successfully resolving 86.3% of user inquiries without external assistance. This surpasses the industry benchmark of 70%–90%, placing it in the “high” category, which indicates effective handling of user interactions and reduced support costs [20].

The chatbot’s performance aligns with industry standards, as highlighted by [20], where a containment rate of 80%–90% is considered a positive sign of efficiency and effectiveness. The BISU Faculty Manual Chatbot’s 86.3% containment rate validates its proficiency in delivering precise and autonomous responses, minimizing the need for human support. This demonstrates its success in enhancing user interaction experiences while maintaining cost efficiency, as supported by benchmarks discussed in [21].

To evaluate user satisfaction, a 5-star rating survey was conducted with 43 participants, focusing on response time, adapt-

ability, and overall performance. Table II shows an average rating of 4 stars, indicating high user satisfaction. This aligns with [22], where chatbot satisfaction ratings ranged from 3 to 5 stars, driven by efficiency, flexibility, and functionality. Additionally, [23] highlights that a well-designed interface and quality service delivery significantly enhance user experience, fostering trust and satisfaction. Future improvements could focus on optimizing response times, refining adaptability to complex queries, and enhancing the interface to further boost satisfaction and reliability.

User's Question	Chatbot Correct Response	Chatbot Incorrect Response	Chatbot Fallback Response
<i>Learning Management System</i>			✓
<i>What is Learning Management System?</i>	✓		
<i>What is Instruction</i>	✓		
<i>What to do if a student have 3 consecutive absences?</i>		✓	
<i>I am currently hired as an instructor in this institution, am I eligible on the vacation service credit</i>	✓		

User No. 023

Please rate our system:

Please respond to the question below, where.

5 star – Very Satisfied
4 star – Satisfied
3 star – Neutral
2 star – Dissatisfied
1 star – Very Dissatisfied

How satisfied are you with the overall user experience of our chatbot system?

★★★★★

Fig. 3. Chatbot Evaluation Form

Figure 3 shows the sample format of the evaluation form utilized by the thesis writers for the manual evaluation of the chatbot’s performance. The form was distributed to users selected through a random sampling method to gather ratings on the system’s effectiveness.

E. Results And Discussions

The evaluation of the BISU Faculty Manual Chatbot was done based on its contextual and accurate response to faculty queries. The chatbot achieved a Containment Rate of 86.3%, which indicates that the majority of user queries were resolved successfully without escalation to human support. The Fallback Rate was 6.1%, which indicates instances where the chatbot was unable to respond appropriately, while the Handoff Rate was 7.6%, which indicates instances where faculty members had to be directed to manual support. These results indicate the reliability of the chatbot in resolving faculty issues autonomously, with minimal errors.

In addition, the teachers tested the chatbot with a total average satisfaction of 4 stars, noting its ease of use, precision, and overall performance in responding to related queries. Use of the TinyLlama generative AI model, which was fine-tuned using the BISU Faculty Manual 2022 Revised Edition,

contributed significantly to the chatbot's capacity to produce accurate responses. Use of Django for frontend and backend programming also provided an easy and friendly interface.

Though having a high rate of containment, there were still some problems encountered, including the handling of complex and unclear questions, which translated to fallback messages. Incorporating the chatbot's dataset and using conversational memory is proposed to further enhance its ability to understand and respond to more complex faculty queries. Overall, the BISU Faculty Manual Chatbot has proven to be an efficient tool in automating organizational policy access of faculty, reducing information search time, and enhancing administrative efficiency as a whole.

V. CONCLUSION

The study concluded that the BISU Faculty Manual Chatbot, built on the Django framework and powered by the TinyLlama model from Hugging Face, effectively integrates frontend and backend functionality to create a user-friendly and efficient system. By replacing manual navigation with an AI-driven approach, the chatbot streamlines access to faculty manual information, reduces operational costs, and enhances institutional efficiency. Its well-optimized architecture ensures accurate, context-relevant responses with minimal resource usage, as confirmed by strong performance metrics and high reliability. While the chatbot has received positive user feedback, further improvements in handling complex queries and refining the interface could enhance its overall effectiveness and user experience.

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